(Please write your Enrollment Number) Enrollment No._____

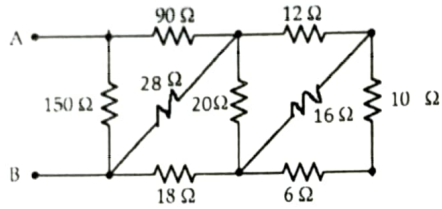
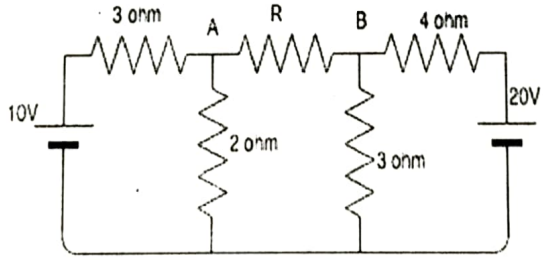
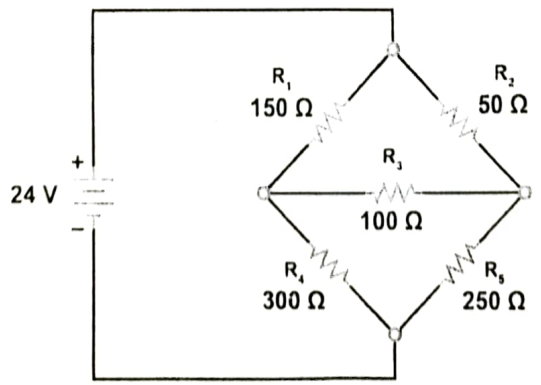
MID-TERM EXAMINATION

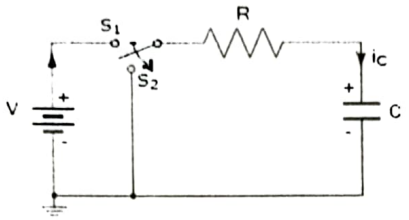
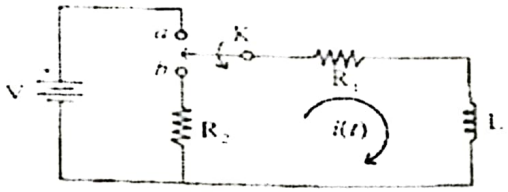
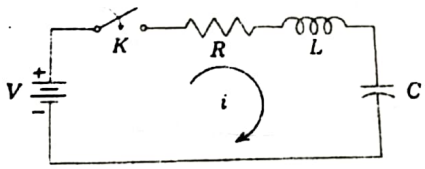
(Course Name: B. Tech.) (Semester: 2)

(March 2025) OFF LINE mode

Subject Code: BEC 104	Subject: Network Analysis and Synthesis
Time : 1 ½Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
	a) Define Kirchhoff's current and voltage law.		CO1
	b) Find the equivalent resistance between the terminal A and B. 		CO2
	c) Define initial value and final value theorem.		CO2
	d) Define unit step, unit ramp and unit impulse signals.		CO1
	UNIT I		CO Mapping
Q2	Attempt any two parts	(5*2=10)	
	a) Derive expression for conversion of resistive delta network to resistive star network.		CO2
	b) Define Maximum Power transfer theorem. What value of resistance R may be connected in the below circuit so that Maximum Power transfer takes place through it? Obtain the value of maximum power transferred. 		CO3
	c) Find the value of current flowing in the resistance 100 Ohm (R_3)? 		CO3

	UNIT II		CO Mapping
Q3	**Attempt any two parts**	**(5*2=10)**	
	a) For the circuit shown below the switch is at S_1 and it achieved steady state. Now at t=0, the switch is moved to position S_2, derive the expression of current flowing in the circuit. 		CO2
	b) In the network shown in figure below, the switch K is moved from position 'b' to 'a' at t=0. Find the Particular solution for Current i(t). 		CO2
	d) In the circuit shown in figure below, V=10V, R=10Ω, L=1H. C=10μFand Vc(0)=0. Find the value of $i(0^+)$, $\frac{di(0^+)}{dt}$ and $\frac{di^2(0^+)}{dt^2}$. 		CO2